

Herald — TELEGRAM

Herald — Telegram Channel Setup Guide

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Field	Value
Status	LIVE (HRD-011 — code complete; awaiting operator credentials for E17/E19 live evidence)
Spec ref	V3 §11.1 + §32.2
HRD	HRD-011
Env vars	HERALD_TGRAM_BOT_TOKEN, HERALD_TGRAM_CHAT_ID, HERALD_TGRAM_LIVE_INBOUND (optional),

Field	Value
	HERALD_TGRAM_WEBHOOK_SECRET (reserved for future webhook ingress)
Code	commons_messaging/channels/ tgram/ (telebot.v3 vendored at submodules/telebot/ v3.3.8)

This guide walks you through every step needed to enable Telegram in Herald — from creating the bot to verifying the live integration test passes. Once you complete these steps, set the env vars in your shell or `.env`, hand the tokens back to the operator, and we will close HRD-011 with live E17/E19 evidence.

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Pre-requisites

- A Telegram account (mobile, desktop, or web client). The account is YOURS; the bot you create is a separate logical entity owned by your account.
- The Herald checkout at `/Users/milosvasic/Projects/Herald` (or wherever you cloned it).
- Either:
 - podman or docker available locally (for the quickstart compose stack), OR
 - a reachable Postgres instance per the §“Postgres” entries in `OPERATOR_CREDENTIALS.md`.

Step 1 — Create a bot via @BotFather

Telegram’s official “create-a-bot” interface is a bot called BotFather. You’ll chat with it to create your bot.

1. Open Telegram → tap the search icon → type BotFather.
2. Tap the result with the verified-blue-checkmark next to its username (the real one is @BotFather). **Don’t** tap any imposter accounts; they cannot create real bots and may phish you.
3. Start the chat with `/start`.

4. Send `/newbot` to BotFather.

5. BotFather asks for a display name. Pick something descriptive — examples:

- Herald Operator Bot
- Herald CI Notifications
- MyProject Notify Bot

6. BotFather asks for a username. Must end in bot (or `_bot`). Examples:

- `herald_operator_bot`
- `myproject_notify_bot`

7. BotFather replies with a **bot token** of the form:

`1234567890:AAH1lab2c3d4e5f6g7h8i9j0kK_LmN3oPqR4S5t`

This token is the bot's password. Treat it like one — never paste it into a PR, a Slack channel, a commit message, or anywhere indexed.

8. **Copy the token immediately** and put it somewhere safe (password manager, secrets vault, secure note).

Token rotation: if you ever leak this token, message `/revoke` to BotFather and pick the bot to invalidate the old token + receive a new one. This is the only fix — git history scrubbing alone does NOT invalidate a leaked Telegram token.

Step 2 — Get the chat ID

Your bot needs to know **where** to send messages. That destination is identified by a numeric **chat ID** — a chat is a one-on-one DM, a group, or a channel.

⚠ **Read Step 2.5 FIRST** if you intend to use a group. Group chat IDs require working around Telegram's Privacy Mode, which is **ON by default** on new bots. Operators routinely waste time wondering why `getUpdates` returns empty until they learn this.

Choose your chat type:

Option A — Direct messages (you `[?]` bot) — works ALWAYS

This path bypasses Privacy Mode entirely. **Use it first** if Step 2.5 below hasn't yet been completed.

1. In Telegram, search for your bot's username (e.g. `@herald_operator_bot`).
2. Tap **Start** to open a DM with it. (The `/start` command is what introduces YOU as a chat the bot knows about.)
3. Send any message (e.g. `hello`).
4. Run the diagnostic script (recommended):

```
bash scripts/tgram_diagnose.sh
```

Or open in a browser:

```
https://api.telegram.org/bot<YOUR-BOT-TOKEN>/getUpdates
```

5. The output shows `chat_id=123456789 type=private name=YourName`.
The id value (positive integer for DMs) is your chat ID.

Option B — Group chat (bot in a group) — see Step 2.5 first

1. Create a Telegram group (or use an existing one).
2. Add your bot to the group via the group's **Add Member** menu.
3. **Complete Step 2.5 below** to ensure the bot can actually see group messages.
4. After completing Step 2.5, send a message inside the group (any text if Privacy Mode is OFF; an @<bot-username> hi mention OR a /command if Privacy Mode is ON).
5. Run `bash scripts/tgram_diagnose.sh` or visit `https://api.telegram.org/bot<TOKEN>/getUpdates`.
6. Find the `chat_id=...` `type=supergroup` (or `group`) block.
7. The id for groups is a **negative number** like `-1001234567890`.

Option C — Channel (one-way broadcast)

Channels are one-way: the bot can post but cannot read messages. Useful for notifications without two-way interaction.

1. Create a Telegram channel.
2. Add your bot as an **Administrator** (a regular member does NOT have post permission). Grant at least “Post Messages”.
3. Send any message to the channel (from your own account, since the bot is admin not subscriber).
4. Run `bash scripts/tgram_diagnose.sh`.
5. Locate `chat_id=...` `type=channel`. Again a negative number.

Confirm the chat ID

Paste the chat ID into a temporary file or note. You'll set it as `HERALD_TGRAM_CHAT_ID` in the next step.

Step 2.5 — Privacy Mode (LOAD-BEARING for groups)

This is the #1 reason new operators get stuck. Skip it and you will waste hours wondering why `getUpdates` returns empty.

When you create a bot via @BotFather, Telegram defaults its **Privacy Mode** to **ON**. With Privacy Mode ON, the bot CAN be added to a group but Telegram filters incoming group messages **before they ever reach the bot's update stream**. The bot only sees:

- Slash commands (/start, /anything)
- Messages that contain @<bot-username> (literal @-mention of the bot's username)
- Replies to messages the bot itself sent
- Service messages (member joins, etc.)

Plain chat messages from group members are INVISIBLE to the bot in Privacy Mode. This is why getUpdates is often empty even though messages clearly exist in the group.

Check your Privacy Mode state

Run:

```
bash scripts/tgram_diagnose.sh
```

The getMe output includes can_read_all_group_messages. If it says False, Privacy Mode is **ON**. If True, it's OFF.

Two paths to fix

Path A — Quick workaround (no settings change)

In the group, send a message that EXPLICITLY contains the bot's username:

```
@atmosphere_worker_bot hi
```

(Substitute your real bot username from getMe.) Telegram delivers this to the bot regardless of Privacy Mode because it's a mention.

After sending, run `bash scripts/tgram_diagnose.sh` again — the chat should appear.

Common gotchas: - Auto-correct on mobile sometimes “corrects” the bot username. Verify exactly: @<your-bot-username> — must match getMe's username field verbatim. - Slash command /start@<bot-username> also works (it's both a command and a mention). - A reply to one of the bot's previous messages works too.

Path B — Disable Privacy Mode in @BotFather (best for production use)

If you want Herald to **route ALL group messages** to Claude Code (not just @-mentions), disable Privacy Mode:

1. Open Telegram → chat with @BotFather
2. Send /mybots
3. Select your bot (e.g. @atmosphere_worker_bot)
4. Tap **Bot Settings**

5. Tap **Group Privacy**
6. Tap **Turn off**
7. BotFather replies: “Done. The bot will see all messages.”
8. **Important — the change takes effect only for new sessions:** REMOVE the bot from the group, then RE-ADD it.
9. Send any plain message in the group.
10. Run `bash scripts/tgram_diagnose.sh` — the chat should appear.

When neither path works

If you’ve completed Path A or B and `getUpdates` is still empty:

- **Verify the bot is actually in the group:** open the group’s member list. If the bot is missing, re-add it.
- **Verify you sent the message AFTER the privacy change** (Path B): privacy changes don’t retroactively re-deliver historic messages.
- **Verify you sent the message AFTER adding the bot** (always): updates from before the bot joined are never delivered.
- **Try a DM /start first** (Option A above): DMs ALWAYS work. If the DM `chat_id` appears but the group `chat_id` doesn’t, the issue is isolated to the group’s privacy configuration.
- **Check if a previous `getUpdates` consumed the update:** Telegram acks updates after each poll. Send a NEW message to force a fresh update into the queue. The diagnostic script uses `offset=-1` (read latest without acking) to mitigate this, but a previously-acked update is gone.

Diagnostic command (canonical helper)

The `scripts/tgram_diagnose.sh` helper runs all three diagnostic API calls — `getMe`, `getWebhookInfo`, `getUpdates` — and reports findings in operator-actionable form:

```
bash scripts/tgram_diagnose.sh
```

Output shape:

```
=== 1. getMe – token validity + Privacy Mode setting ===
bot username:          @your_bot
bot id:                1234567890
can_read_all_group_messages: False
! Privacy Mode is ON.
  - commands (/...)
  - @-mentions of itself
  - replies to its own messages
  Plain chat messages from group members are NOT delivered.

=== 2. getWebhookInfo – confirm getUpdates is the live channel ===
no webhook configured (good)

=== 3. getUpdates – list of chats the bot has received updates
from ===
updates received: 1
```

```
chat_id=-1001234567890 type=supergroup name=Herald
Notifications source=message
```

If updates received: 0 with explanation, follow the suggested fix.

Step 3 — Provide the credentials to Herald

Choose ONE of these two paths (or both — the resolution order in OPERATOR_CREDENTIALS.md handles overlap correctly).

Path A — Shell-export (recommended for personal workstation)

Add to your ~/.zshrc (macOS default) or ~/.bashrc (Linux default):

```
# Herald — Telegram (HRD-011)
export HERALD_TGRAM_BOT_TOKEN='1234567890:AAH1lab2c3d4e5f6g7h8i9j0kK_LmN3oPqR4S5t'
export HERALD_TGRAM_CHAT_ID='-1001234567890'      # or your DM chat
ID
```

Then either run source ~/.zshrc (or ~/.bashrc) or open a new terminal.

Verify:

```
echo "${HERALD_TGRAM_BOT_TOKEN:0:10}..."    # should print the
first 10 chars
echo "${HERALD_TGRAM_CHAT_ID}"
# should print the chat ID
```

Path B — Project-local .env

Edit (or create) /Users/milosvasic/Projects/Herald/.env:

```
HERALD_TGRAM_BOT_TOKEN=1234567890:AAH1lab2c3d4e5f6g7h8i9j0kK_LmN3oPqR4S5t
HERALD_TGRAM_CHAT_ID=-1001234567890
```

Set restrictive permissions:

```
chmod 600 .env
```

For native pherald (NOT through compose), source it into your shell first:

```
set -a; source .env; set +a
```

For compose runs, docker-compose reads .env automatically.

Per §11.4.10: never git add .env. The repo's .gitignore already covers it (line 28), but always double-check before committing.

Step 4 — Verify HealthCheck

The HealthCheck integration test issues a real `getMe` call against the Bot API. It proves the token is valid and the bot is enabled.

```
cd /Users/milosvasic/Projects/Herald
go test ./commons_messaging/channels/tgram/ -tags=integration
    -run TestHealthCheck_LiveBotAPI -count=1 -timeout=60s
```

Expected:

```
--- PASS: TestHealthCheck_LiveBotAPI (1.x s)
PASS
```

If you see `SKIP`, the env vars aren't visible to `go test`. Re-source your shell or run the command with the env vars inline:

```
HERALD_TGRAM_BOT_TOKEN='...' HERALD_TGRAM_CHAT_ID='...' \
go test ./commons_messaging/channels/tgram/ -tags=integration
    -run TestHealthCheck_LiveBotAPI -count=1 -timeout=60s
```

If you see `FAIL` with `getMe` returned empty body or HTTP errors, the token is invalid. Re-check Step 1.

Step 5 — Verify Send (E17)

The E17 evidence test sends a real Telegram message to your configured chat AND persists a row in the `outbound_delivery_evidence` table. It proves the full Send + persistence round-trip works.

Pre-requisite: container runtime (podman or docker) available. The test boots a Postgres container automatically.

```
cd /Users/milosvasic/Projects/Herald
go test ./commons_messaging/channels/tgram/ -tags=integration
    -run TestSend_PersistsDeliveryEvidence -count=1
    -timeout=300s -v
```

Expected:

```
--- PASS: TestSend_PersistsDeliveryEvidence (Xs)
PASS
```

You should also see the test message arrive in your Telegram chat — something like:
> Herald E17 persist test 2026-05-21T11:23:45.789Z

The §107 anti-bluff guard asserts the persisted `channel_message_id` equals the Telegram-side `MessageID` exactly — not a Herald-generated UUID. A spoofed adapter that returned a fake ID would fail this test.

Step 6 — (Optional) Verify Subscribe + Vertical Slice (E19)

E19 is the canonical Telegram → Claude Code → Telegram round-trip. It requires:

1. All of Step 3's env vars set.
2. HERALD_CLAUDE_BIN resolves to a working `claude` CLI (default: looked up via `$PATH`).
3. HERALD_CLAUDE_PROJECT_NAME set (see [../dispatchers/CLAUDE_CODE.md](#)).
4. HERALD_TGRAM_LIVE_INBOUND=1 to enable the live-inbound code path.
5. **You hand-send a Telegram message to the bot during the test window** (60s default).

```
HERALD_TGRAM_LIVE_INBOUND=1 \
go test ./commons_messaging/ -tags=integration -run
    TestVerticalSlice_TelegramClaudeRoundTrip -count=1
    -timeout=300s -v
```

Within the 60s window: open Telegram → send any message to your bot. The handler will (a) receive your message, (b) dispatch it to Claude Code, (c) parse the reply, (d) send the reply back to your chat.

Expected:

```
E19 PASS: tenant=...; inbound=1; dispatch ok; outbound ok
(channel_id=...)
--- PASS: TestVerticalSlice_TelegramClaudeRoundTrip (Xs)
```

You should see TWO messages in your Telegram chat: yours, then Claude's reply.

Step 7 — (Future) Webhook ingress for production

Long-poll (Steps 1-6) is what Herald uses today. For lower latency in production, Telegram supports webhook ingress — Telegram POSTs each update to your endpoint.

Status: NOT YET IMPLEMENTED — reserved for HRD-NNN.

When webhook ingress lands, you'll:

1. Generate a 256-bit hex secret:

```
openssl rand -hex 32
```

2. Set HERALD_TGRAM_WEBHOOK_SECRET=<the-hex>.

3. Configure the bot's webhook URL via the Bot API:

```
https://api.telegram.org/bot<TOKEN>/setWebhook?url=https://
your-herald.example.com/v1/webhooks/tgram&secret_token=<the-
hex>
```

4. Telegram POSTs each update to `/v1/webhooks/tgram` with the secret in the `X-Telegram-Bot-API-Secret-Token` header for verification.

Until webhook ingress lands, leave `HERALD_TGRAM_WEBHOOK_SECRET` unset and rely on long-poll.

Troubleshooting

“401 Unauthorized” from `getMe`: Token is wrong. Re-copy from BotFather; check no trailing whitespace.

`getUpdates` returns 409 Conflict: A webhook is configured for this bot. Either use the webhook path (Step 7) OR delete the webhook to switch back to long-poll:

`https://api.telegram.org/bot<TOKEN>/deleteWebhook`

Bot doesn’t see group messages: For groups, you must DISABLE “Privacy Mode” in BotFather: 1. Chat with `@BotFather` → `/mybots` → select your bot → “Bot Settings” → “Group Privacy” → “Turn off”. 2. Re-add the bot to the group if it was already added (the privacy change takes effect for new sessions).

Test SKIPS despite env vars set: Some test runners scope env vars per-process. Run with inline env: `HERALD_TGRAM_BOT_TOKEN='...' go test ....`

MarkdownV2 parse error: Herald uses MarkdownV2 parse mode by default (spec §11.1). Telegram requires specific escaping for `_`, `*`, `[`, `]`, `(`, `)`, `~`, ```, `>`, `#`, `+`, `-`, `=`, `|`, `{`, `}`, `.`, `!`. If a message contains unescaped versions, the Send returns 400. Use `commons.Body.Plain` for plain-text without parse mode interpretation.

“chat not found”: The bot was removed from the group, or you’re using the wrong chat ID. Re-run `getUpdates` to confirm the chat exists.

Spec + code references

- **Spec:** V3 §11.1 (Telegram capabilities + Bot API surface), §32.2 (subscriber-reply long-poll loop 25s + 30s safety-net)
- **HRD:** HRD-011 (Telegram channel adapter live integration)
- **Code:**
 - `commons_messaging/channels/tgram/tgram.go` — Adapter struct + URL parser + Capabilities
 - `commons_messaging/channels/tgram/healthcheck.go` — `getMe` `HealthCheck`
 - `commons_messaging/channels/tgram/send.go` — `Bot.Send` (sendMessage MarkdownV2 + forum-topic ThreadID)
 - `commons_messaging/channels/tgram/subscribe.go` — LongPoller 25s + 30s safety-net per §32.2
 - `commons_messaging/channels/tgram/persist.go` — `NewWithStorage` + `SendForTenant` (outbound_delivery_evidence persistence)
- **Tests:**
 - `commons_messaging/channels/tgram/healthcheck_integration_test.go` — E17 prerequisite

- `commons_messaging/channels/tgram/send_integration_test.go` — E17 (live Telegram + non-empty `Receipt.ChannelMsgID`)
- `commons_messaging/channels/tgram/subscribe_integration_test.go` — E19 inbound half
- `commons_messaging/channels/tgram/persist_integration_test.go` — E17 persistence (exact `channel_message_id` match)
- `commons_messaging/vertical_slice_integration_test.go` — E19 full slice
- **Vendored SDK:** `submodules/telebot/` (`gopkg.in/telebot.v3` at v3.3.8, pinned to keep `gopkg.in/telebot.v3` import-path stable per §11.4.74)
- **Related:** `OPERATOR_CREDENTIALS.md` §“Telegram”; `../dispatchers/CLAUDE_CODE.md` for the LLM dispatch half of the vertical slice